

L2 Geometry Workbook for Profile Sieve EL

Three-Part Proof: Geometry, Small Stochastic Bounds, and Merging

Solved derivation workbook

Contents

I	Geometric and Convex KKT Body	3
1	Model Setup: Full Geometry in $L^2(P_T)$ and $L^2(P)$	3
2	Finite-Sample Hilbert Space: Viewing the Entire Model in $L^2(P_T)$	4
2.1	The empirical space	4
2.2	The full finite-sample model	4
2.3	Candidate nuisance subspaces	4
2.4	Orthogonal projection	6
3	Profiling Is Orthogonal Projection	7
3.1	Break search as subspace selection	8
4	Residualized Score Direction	9
5	Known True Partition: Wilks Logic in Geometry	10
6	Omitted Breaks: False Predictivity as a Projection Drift	10
7	Estimated Partition	12
8	EL as a Finite-Sample Convex KKT Problem	12
9	Population Hilbert Space	14
9.1	Population object in $L^2(P)$	14
9.2	Population nuisance space	14
10	Population Identification	14
II	Small Stochastic Bounds	15
11	Small Stochastic Bounds	15
III	Merging Part	16

12 Merging Part: Moment Bridge, EL KKT, and χ^2_1	17
12.1 Merge A: consistency	17
12.2 Merge B: Wilks inference	17
13 Three-Part Logic Checklist	19
14 Minimal Final Statement	19

Part I

Geometric and Convex KKT Body

1 Model Setup: Full Geometry in $L^2(P_T)$ and $L^2(P)$

The purpose of this workbook is to rewrite the whole proof logic as Hilbert space geometry. The sample space is not first probability; it is first the finite empirical Hilbert space

$$L^2(P_T) = (\mathbb{R}^T, \langle a, b \rangle_T = T^{-1}a'b).$$

The population limit is the Hilbert space

$$L^2(P), \quad \langle f, g \rangle_P = \mathbb{E}\{fg\}.$$

The model setup is the following chain:

$\mathbf{y} = \beta_0 \mathbf{x} + \boldsymbol{\mu}_0 + \mathbf{u}$	model in $L^2(P_T)$,
$\mu_{0,t} = m_{j_0(t)}(W_{t-1}), \quad j_0(t) = j \iff k_{j,0} < t \leq k_{j+1,0}$	true nuisance vector,
$\mathcal{N}_{K,\Lambda,T} = \text{col}(\mathbf{Q}_{K,\Lambda})$	nuisance subspace,
$\mathbf{M}_{K,\Lambda} = I - \Pi_{K,\Lambda,T}$	residual-maker,
$S_T(\beta, \Lambda) = \sqrt{T} \langle \mathbf{M}_{K,\Lambda}(\mathbf{y} - \beta \mathbf{x}), \mathbf{M}_{K,\Lambda} \mathbf{w} \rangle_T$	projected score.

Here $\boldsymbol{\mu}_0$ is exactly the stacked nonparametric nuisance part of the model. If there are no nuisance breaks, $\mu_{0,t} = m(W_{t-1})$. With breaks, $\mu_{0,t} = m_j(W_{t-1})$ on regime j , so $\boldsymbol{\mu}_0$ is not one stable function evaluated over the whole sample; it is the true piecewise nuisance surface evaluated observation by observation.

EL then uses the scalar products

$$Z_t(\beta, \Lambda) = \{\mathbf{M}_{K,\Lambda}(\mathbf{y} - \beta \mathbf{x})\}_t \{\mathbf{M}_{K,\Lambda} \mathbf{w}\}_t$$

to form a one-dimensional empirical likelihood. EL is not the Hilbert projection; EL is the likelihood wrapper around the projected score.

Three-part proof map. Part I is finite-sample geometry: profiling, residual scores, and EL are all KKT/convex objects in $L^2(P_T)$. Part II is small stochastic verification: Gram stability, sieve and break errors, CLT, variance, max-score, and convex-hull size. Part III merges them: consistency comes from the moment bridge plus unique population zero, while Wilks comes from the EL KKT quadratic reduction plus the standardized score CLT.

2 Finite-Sample Hilbert Space: Viewing the Entire Model in $L^2(P_T)$

2.1 The empirical space

Let

$$\Omega_T = \{1, \dots, T\}, \quad P_T = T^{-1} \sum_{t=1}^T \delta_t.$$

Every sample vector $\mathbf{a} = (a_1, \dots, a_T)'$ is a function $a_T : \Omega_T \rightarrow \mathbb{R}$, with $a_T(t) = a_t$. The empirical inner product is

$$\langle \mathbf{a}, \mathbf{b} \rangle_T = \int a_T b_T dP_T = T^{-1} \sum_{t=1}^T a_t b_t = T^{-1} \mathbf{a}' \mathbf{b}.$$

Thus all variables in the sample are elements of $L^2(P_T)$:

$$\mathbf{y}, \mathbf{x}, \mathbf{w}, \mathbf{u}, \boldsymbol{\mu}_0 \in L^2(P_T).$$

2.2 The full finite-sample model

The break-aware model is the vector identity in $L^2(P_T)$

$$\mathbf{y} = \beta_0 \mathbf{x} + \boldsymbol{\mu}_0 + \mathbf{u}, \tag{2.1}$$

where

$$\mu_{0,t} = \sum_{j=0}^{q_0} m_j(W_{t-1}) \mathbf{1}\{k_{j,0} < t \leq k_{j+1,0}\}.$$

So the full model says: $\mathbf{y}$ lies in the affine set

$$\beta_0 \mathbf{x} + \boldsymbol{\mu}_0 + \mathbf{u}.$$

The inferential problem is to remove $\boldsymbol{\mu}_0$, not by estimating it as a main parameter, but by projecting away its nuisance subspace.

2.3 Candidate nuisance subspaces

For a candidate partition $\Lambda = (k_1, \dots, k_q)$, define the block sieve matrix

$$\mathbf{Q}_{K,\Lambda} = [\mathbf{D}_0(\Lambda) \mathbf{P}, \dots, \mathbf{D}_q(\Lambda) \mathbf{P}].$$

The sample nuisance subspace is

$$\mathcal{N}_{K,\Lambda,T} = \text{col}(\mathbf{Q}_{K,\Lambda}) \subset L^2(P_T). \tag{2.2}$$

An element of this space has the form

$$g_T(t) = \sum_{j=0}^q \mathbf{1}\{t \in I_j(\Lambda)\} \mathbf{P}_K(W_{t-1})' c_j.$$

This is the finite-sample version of a regime-specific nonparametric nuisance function.

Let $j_\Lambda(t)$ denote the regime index induced by Λ . For a regime-wise nuisance tuple $h = (h_0, \dots, h_q)$, define its stacked sample vector by

$$\mathbf{h}_{\Lambda,T} = (h_{j_\Lambda(1)}(W_0), \dots, h_{j_\Lambda(T)}(W_{T-1}))'.$$

This vector is the empirical version of the candidate nuisance direction.

Lemma 1 (Uniform finite-sieve approximation in $L^2(P_T)$). *Fix (K, Λ, T) and a nuisance class $\mathcal{H}$. For each $h \in \mathcal{H}$, let $\mathbf{h}_{\Lambda,T} \in L^2(P_T)$ be the stacked nuisance vector defined above. Define the finite-sieve approximation radius*

$$\rho_{K,T}(\mathcal{H}, \Lambda) = \sup_{h \in \mathcal{H}} \inf_{\gamma \in \mathbb{R}^{(q+1)K}} \|\mathbf{h}_{\Lambda,T} - \mathbf{Q}_{K,\Lambda} \gamma\|_T. \quad (2.3)$$

If $\rho_{K,T}(\mathcal{H}, \Lambda) = o_p(1)$, then the desired uniform $L^2(P_T)$ approximation holds:

$$\sup_{h \in \mathcal{H}} \|\mathbf{h}_{\Lambda,T} - \mathbf{Q}_{K,\Lambda} \gamma_{K,\Lambda}(h)\|_T = \sup_{h \in \mathcal{H}} \|\mathbf{r}_{K,\Lambda,T}(h)\|_T = \rho_{K,T}(\mathcal{H}, \Lambda) = o_p(1). \quad (2.4)$$

Here, for each h ,

$$\gamma_{K,\Lambda}(h) \in \operatorname{argmin}_{\gamma \in \mathbb{R}^{(q+1)K}} \|\mathbf{h}_{\Lambda,T} - \mathbf{Q}_{K,\Lambda} \gamma\|_T^2,$$

$$\mathbf{h}_{K,\Lambda,T} = \mathbf{Q}_{K,\Lambda} \gamma_{K,\Lambda}(h), \quad \mathbf{r}_{K,\Lambda,T}(h) = \mathbf{h}_{\Lambda,T} - \mathbf{h}_{K,\Lambda,T}.$$

The ambient space is $L^2(P_T) \cong \mathbb{R}^T$, so $\mathbf{h}_{\Lambda,T}$, $\mathbf{h}_{K,\Lambda,T}$, and $\mathbf{r}_{K,\Lambda,T}(h)$ are all $T \times 1$ vectors. The design matrix and coefficient vector have dimensions

$$\mathbf{Q}_{K,\Lambda} \in \mathbb{R}^{T \times (q+1)K}, \quad \gamma_{K,\Lambda}(h) \in \mathbb{R}^{(q+1)K}.$$

The finite nuisance space has dimension

$$d_{K,\Lambda,T} = \operatorname{rank}(\mathbf{Q}_{K,\Lambda}) \leq (q+1)K,$$

and

$$\mathbf{h}_{\Lambda,T} = \mathbf{Q}_{K,\Lambda} \gamma_{K,\Lambda}(h) + \mathbf{r}_{K,\Lambda,T}(h), \quad (2.5)$$

where

$$\mathbf{Q}_{K,\Lambda} \gamma_{K,\Lambda}(h) \in \mathcal{N}_{K,\Lambda,T} \subset L^2(P_T), \quad \mathbf{r}_{K,\Lambda,T}(h) \in \mathcal{N}_{K,\Lambda,T}^{\perp_T} \subset L^2(P_T).$$

Thus $\mathbf{r}_{K,\Lambda,T}(h)$ is a $T \times 1$ approximation-residual vector living in a $(T - d_{K,\Lambda,T})$ -dimensional orthogonal complement, and

$$\mathbf{Q}_{K,\Lambda}' \mathbf{r}_{K,\Lambda,T}(h) = 0.$$

Proof. The column space $\mathcal{N}_{K,\Lambda,T} = \text{col}(\mathbf{Q}_{K,\Lambda})$ is a finite-dimensional closed subspace of $L^2(P_T)$. Therefore the Hilbert projection theorem gives the closest finite-sieve approximation to every $\mathbf{h}_{\Lambda,T}$:

$$\Pi_{K,\Lambda,T} \mathbf{h}_{\Lambda,T} \in \mathcal{N}_{K,\Lambda,T}.$$

Because every element of $\mathcal{N}_{K,\Lambda,T}$ is $\mathbf{Q}_{K,\Lambda} \gamma$, the projection can be written as

$$\Pi_{K,\Lambda,T} \mathbf{h}_{\Lambda,T} = \mathbf{Q}_{K,\Lambda} \gamma_{K,\Lambda}(h)$$

for any least-squares minimizer $\gamma_{K,\Lambda}(h)$. If $\mathbf{Q}_{K,\Lambda}$ is rank deficient, the coefficient vector need not be unique, but the fitted vector is unique.

Define

$$\mathbf{r}_{K,\Lambda,T}(h) = \mathbf{h}_{\Lambda,T} - \Pi_{K,\Lambda,T} \mathbf{h}_{\Lambda,T}.$$

The projection theorem gives

$$\langle \mathbf{r}_{K,\Lambda,T}(h), \mathbf{v} \rangle_T = 0 \quad \text{for every } \mathbf{v} \in \mathcal{N}_{K,\Lambda,T},$$

which implies $\mathbf{Q}'_{K,\Lambda} \mathbf{r}_{K,\Lambda,T}(h) = 0$. This proves the orthogonal decomposition (2.5) and the location $\mathbf{r}_{K,\Lambda,T}(h) \in \mathcal{N}_{K,\Lambda,T}^\perp$.

By the definition of orthogonal projection,

$$\|\mathbf{r}_{K,\Lambda,T}(h)\|_T = \inf_{\gamma \in \mathbb{R}^{(q+1)K}} \|\mathbf{h}_{\Lambda,T} - \mathbf{Q}_{K,\Lambda} \gamma\|_T.$$

Taking the supremum over $h \in \mathcal{H}$ gives exactly (2.4). Hence the mathematical task behind this lemma is to verify $\rho_{K,T}(\mathcal{H}, \Lambda) = o_p(1)$ from the chosen basis, smoothness class, and growth rate of K . An arbitrary nuisance direction is not assumed to be exactly spanned by the sieve basis; it is approximated by its finite-sieve projection, and the leftover is $\mathbf{r}_{K,\Lambda,T}(h)$. $\square$

2.4 Orthogonal projection

Let $\Pi_{K,\Lambda,T}$ be the $L^2(P_T)$ -orthogonal projection onto $\mathcal{N}_{K,\Lambda,T}$. In coordinates,

$$\Pi_{K,\Lambda,T} = \mathbf{Q}_{K,\Lambda} (\mathbf{Q}'_{K,\Lambda} \mathbf{Q}_{K,\Lambda})^\dagger \mathbf{Q}'_{K,\Lambda}.$$

The residual-maker is

$$\mathbf{M}_{K,\Lambda} = I - \Pi_{K,\Lambda,T}. \tag{2.6}$$

It satisfies the projection identities

$$\mathbf{M}'_{K,\Lambda} = \mathbf{M}_{K,\Lambda}, \quad \mathbf{M}_{K,\Lambda}^2 = \mathbf{M}_{K,\Lambda}, \quad \mathbf{Q}'_{K,\Lambda} \mathbf{M}_{K,\Lambda} = 0.$$

Lemma 2 (Projection identities for the residual-maker). *Let $\mathbf{Q} = \mathbf{Q}_{K,\Lambda}$, $\mathbf{G} = \mathbf{Q}'\mathbf{Q}$,*

$$\Pi = \mathbf{Q} \mathbf{G}^\dagger \mathbf{Q}', \quad \mathbf{M} = I - \Pi,$$

where $\mathbf{G}^\dagger$ is the Moore-Penrose inverse. Then Π and $\mathbf{M}$ are symmetric and idempotent, and $\mathbf{M}$ annihilates the sieve nuisance space:

$$\mathbf{M}' = \mathbf{M}, \quad \mathbf{M}^2 = \mathbf{M}, \quad \mathbf{M}\mathbf{Q} = 0, \quad \mathbf{Q}'\mathbf{M} = 0.$$

Proof. Since $G = Q'Q$ is symmetric, its Moore-Penrose inverse $G^\dagger$ is also symmetric. Hence

$$\Pi' = \{QG^\dagger Q'\}' = QG^\dagger Q' = \Pi.$$

For idempotence, use the Moore-Penrose identity $G^\dagger GG^\dagger = G^\dagger$:

$$\Pi^2 = QG^\dagger Q'QG^\dagger Q' = QG^\dagger GG^\dagger Q' = QG^\dagger Q' = \Pi.$$

Therefore

$$M' = I - \Pi' = M, \quad M^2 = (I - \Pi)^2 = I - 2\Pi + \Pi^2 = I - \Pi = M.$$

It remains to show that M kills the columns of Q . Since $\mathcal{R}(G) = \mathcal{R}(Q')$ and $G^\dagger G$ is the orthogonal projector onto $\mathcal{R}(G)$,

$$QG^\dagger G = Q.$$

Thus

$$\Pi Q = QG^\dagger Q'Q = QG^\dagger G = Q, \quad MQ = (I - \Pi)Q = 0.$$

Finally, because M is symmetric, $Q'M = (MQ)' = 0$. This proves all projection identities. $\square$

3 Profiling Is Orthogonal Projection

For a trial value β , define the partial outcome

$$\mathbf{y}_\beta^\circ = \mathbf{y} - \beta \mathbf{x}.$$

It is the original outcome $\mathbf{y}$ after subtracting the trial linear component $\beta \mathbf{x}$.

Profiling the nuisance means finding the closest nuisance vector to the partial outcome $\mathbf{y}_\beta^\circ$ in $\mathcal{N}_{K,\Lambda,T}$:

$$\hat{\boldsymbol{\mu}}_{\beta,\Lambda} = \arg \min_{\mathbf{g} \in \mathcal{N}_{K,\Lambda,T}} \|\mathbf{y}_\beta^\circ - \mathbf{g}\|_T^2.$$

By Hilbert projection,

$$\hat{\boldsymbol{\mu}}_{\beta,\Lambda} = \Pi_{K,\Lambda,T} \mathbf{y}_\beta^\circ, \quad \hat{\mathbf{u}}(\beta, \Lambda) = \mathbf{y}_\beta^\circ - \hat{\boldsymbol{\mu}}_{\beta,\Lambda} = \mathbf{M}_{K,\Lambda} \mathbf{y}_\beta^\circ.$$

Lemma 3 (Profile normal equation). *For every $\mathbf{h} \in \mathcal{N}_{K,\Lambda,T}$,*

$$\langle \hat{\mathbf{u}}(\beta, \Lambda), \mathbf{h} \rangle_T = 0.$$

Equivalently,

$$\mathbf{Q}'_{K,\Lambda} \hat{\mathbf{u}}(\beta, \Lambda) = 0.$$

Proof. This is the KKT condition for a least-squares projection in $L^2(P_T)$. If $\mathbf{h} = \mathbf{Q}_{K,\Lambda} \mathbf{a}$, then

$$\langle \hat{\mathbf{u}}, \mathbf{h} \rangle_T = T^{-1} \hat{\mathbf{u}}' \mathbf{Q}_{K,\Lambda} \mathbf{a} = T^{-1} (\mathbf{Q}'_{K,\Lambda} \hat{\mathbf{u}})' \mathbf{a} = 0.$$

$\square$

3.1 Break search as subspace selection

For fixed β , the partial outcome is

$$\mathbf{y}_\beta^\circ = \mathbf{y} - \beta \mathbf{x}.$$

For a candidate partition $\Lambda = (k_1, \dots, k_q)$, set $k_0 = 0$, $k_{q+1} = T$, and

$$I_j(\Lambda) = \{t : k_j < t \leq k_{j+1}\}.$$

The block sieve matrix is

$$\mathbf{Q}_{K,\Lambda} = [\mathbf{D}_0(\Lambda)\mathbf{P}, \dots, \mathbf{D}_q(\Lambda)\mathbf{P}].$$

For $c = (c'_0, \dots, c'_q)'$, the fitted nuisance vector is $\mathbf{Q}_{K,\Lambda}c$, with component

$$(\mathbf{Q}_{K,\Lambda}c)_t = \mathbf{P}_K(W_{t-1})'c_j \quad \text{if } t \in I_j(\Lambda).$$

Thus minimizing over c is exactly the same as choosing a separate sieve coefficient vector c_j on each segment.

Lemma 4 (Profile RSS is a Hilbert distance). *For fixed (β, Λ) , define the segment cost*

$$C_K(s, e; \beta) = \min_{c \in \mathbb{R}^K} \sum_{t=s}^e \{y_t - \beta x_t - \mathbf{P}_K(W_{t-1})'c\}^2,$$

and the profile RSS

$$RSS_K(\beta, \Lambda) = \sum_{j=0}^q C_K(k_j + 1, k_{j+1}; \beta).$$

Then

$$\begin{aligned} RSS_K(\beta, \Lambda) &= \min_{c \in \mathbb{R}^{(q+1)K}} \|\mathbf{y}_\beta^\circ - \mathbf{Q}_{K,\Lambda}c\|_2^2 \\ &= \inf_{\mathbf{g} \in \mathcal{N}_{K,\Lambda,T}} \|\mathbf{y}_\beta^\circ - \mathbf{g}\|_2^2 \\ &= T \text{dist}_T^2(\mathbf{y}_\beta^\circ, \mathcal{N}_{K,\Lambda,T}) \\ &= T \|\mathbf{M}_{K,\Lambda} \mathbf{y}_\beta^\circ\|_T^2. \end{aligned} \tag{3.1}$$

Proof. Because each segment $I_j(\Lambda)$ has its own coefficient vector c_j ,

$$\begin{aligned} RSS_K(\beta, \Lambda) &= \min_{c_0, \dots, c_q} \sum_{j=0}^q \sum_{t \in I_j(\Lambda)} \{y_t - \beta x_t - \mathbf{P}_K(W_{t-1})'c_j\}^2 \\ &= \min_{c \in \mathbb{R}^{(q+1)K}} \|\mathbf{y} - \beta \mathbf{x} - \mathbf{Q}_{K,\Lambda}c\|_2^2 \\ &= \min_{c \in \mathbb{R}^{(q+1)K}} \|\mathbf{y}_\beta^\circ - \mathbf{Q}_{K,\Lambda}c\|_2^2. \end{aligned}$$

Since $\mathcal{N}_{K,\Lambda,T} = \text{col}(\mathbf{Q}_{K,\Lambda})$, every $\mathbf{g} \in \mathcal{N}_{K,\Lambda,T}$ can be written as $\mathbf{g} = \mathbf{Q}_{K,\Lambda}c$. Therefore

$$\min_c \|\mathbf{y}_\beta^\circ - \mathbf{Q}_{K,\Lambda}c\|_2^2 = \inf_{\mathbf{g} \in \mathcal{N}_{K,\Lambda,T}} \|\mathbf{y}_\beta^\circ - \mathbf{g}\|_2^2.$$

The Euclidean norm and the empirical Hilbert norm satisfy $\|\mathbf{a}\|_2^2 = T\|\mathbf{a}\|_T^2$. Hence

$$\inf_{\mathbf{g} \in \mathcal{N}_{K,\Lambda,T}} \|\mathbf{y}_\beta^\circ - \mathbf{g}\|_2^2 = T \operatorname{dist}_T^2(\mathbf{y}_\beta^\circ, \mathcal{N}_{K,\Lambda,T}).$$

Finally, $\mathbf{M}_{K,\Lambda}$ is the residual-maker for the orthogonal projection onto $\mathcal{N}_{K,\Lambda,T}$, so

$$\operatorname{dist}_T^2(\mathbf{y}_\beta^\circ, \mathcal{N}_{K,\Lambda,T}) = \|\mathbf{M}_{K,\Lambda} \mathbf{y}_\beta^\circ\|_T^2.$$

This proves (3.1). □

Consequently, for known order q , the break estimator is

$$\begin{aligned} \hat{\Lambda}_q(\beta) &= \arg \min_{\Lambda \in \mathcal{L}_q(\epsilon)} RSS_K(\beta, \Lambda) \\ &= \arg \min_{\Lambda \in \mathcal{L}_q(\epsilon)} \operatorname{dist}_T^2(\mathbf{y}_\beta^\circ, \mathcal{N}_{K,\Lambda,T}). \end{aligned}$$

Unknown order adds a penalty to the same distance:

$$\hat{q}(\beta) = \arg \min_{0 \leq q \leq \bar{q}} \left\{ T \operatorname{dist}_T^2(\mathbf{y}_\beta^\circ, \mathcal{N}_{K,\hat{\Lambda}_q(\beta),T}) + \operatorname{pen}_T(q, K) \right\}.$$

Thus break search is subspace selection: each candidate Λ creates a candidate nuisance subspace $\mathcal{N}_{K,\Lambda,T}$, and RSS picks the subspace closest to the partial outcome $\mathbf{y}_\beta^\circ$.

4 Residualized Score Direction

The raw score weight $\mathbf{w} \in L^2(P_T)$ may contain nuisance directions. The geometric score direction is its nuisance-orthogonal component

$$\mathbf{v}_\Lambda = \mathbf{w}^c(\Lambda) = \mathbf{M}_{K,\Lambda} \mathbf{w}.$$

Then

$$\mathbf{v}_\Lambda \in \mathcal{N}_{K,\Lambda,T}^\perp, \quad \langle \mathbf{v}_\Lambda, \mathbf{h} \rangle_T = 0 \quad \forall \mathbf{h} \in \mathcal{N}_{K,\Lambda,T}.$$

The sample profiled moment is the Hilbert inner product

$$\Psi_T(\beta, \Lambda) = \langle \mathbf{M}_{K,\Lambda}(\mathbf{y} - \beta \mathbf{x}), \mathbf{M}_{K,\Lambda} \mathbf{w} \rangle_T. \quad (4.1)$$

The scaled score is

$$S_T(\beta, \Lambda) = \sqrt{T} \Psi_T(\beta, \Lambda).$$

Lemma 5 (Finite-sample nuisance cancellation). *If $\mathbf{g} \in \mathcal{N}_{K,\Lambda,T}$, then*

$$\langle \mathbf{M}_{K,\Lambda} \mathbf{g}, \mathbf{M}_{K,\Lambda} \mathbf{w} \rangle_T = 0.$$

Proof. Because $\mathbf{g} \in \mathcal{N}_{K,\Lambda,T}$, $\mathbf{M}_{K,\Lambda} \mathbf{g} = 0$. Hence the inner product is zero. Equivalently, $(\mathbf{Q}c)' \mathbf{M}_{K,\Lambda} \mathbf{w} = 0$ for every c . □

5 Known True Partition: Wilks Logic in Geometry

Set $\Lambda = \Lambda_0$, and abbreviate

$$\mathbf{Q}_0 = \mathbf{Q}_{K, \Lambda_0}, \quad \mathbf{M}_0 = \mathbf{M}_{K, \Lambda_0}, \quad \mathbf{v}_0 = \mathbf{M}_0 \mathbf{w}.$$

Decompose the true nuisance into its finite-sieve projection and approximation residual:

$$\boldsymbol{\mu}_0 = \mathbf{Q}_0 \mathbf{c}_0 + \mathbf{r}_0, \quad \mathbf{Q}_0 \mathbf{c}_0 \in \mathcal{N}_{K, \Lambda_0, T}, \quad \mathbf{r}_0 \in \mathcal{N}_{K, \Lambda_0, T}^{\perp T}.$$

Lemma 6 (True-partition score decomposition). *At $\beta = \beta_0$ and $\Lambda = \Lambda_0$, the projected score satisfies*

$$S_T(\beta_0, \Lambda_0) = T^{-1/2} \mathbf{u}' \mathbf{v}_0 + T^{-1/2} \mathbf{r}_0' \mathbf{v}_0. \quad (5.1)$$

The first term is the stochastic score. The second term is the sieve approximation-bias term, and this is the term controlled by approximation rates:

$$T^{-1/2} \mathbf{r}_0' \mathbf{v}_0 = T^{-1/2} \mathbf{r}_0' \mathbf{M}_0 \mathbf{w} = o_p(1). \quad (5.2)$$

Proof. At the true parameter, the partial outcome is

$$\mathbf{y}_{\beta_0}^\circ = \mathbf{y} - \beta_0 \mathbf{x} = \boldsymbol{\mu}_0 + \mathbf{u} = \mathbf{Q}_0 \mathbf{c}_0 + \mathbf{r}_0 + \mathbf{u}.$$

Apply the true-partition residual-maker $\mathbf{M}_0$. Since $\mathbf{Q}_0 \mathbf{c}_0$ is in $\mathcal{N}_{K, \Lambda_0, T}$, Lemma 2 gives $\mathbf{M}_0 \mathbf{Q}_0 \mathbf{c}_0 = 0$. Therefore

$$\mathbf{M}_0 \mathbf{y}_{\beta_0}^\circ = \mathbf{M}_0 \mathbf{u} + \mathbf{M}_0 \mathbf{r}_0.$$

The projected score is

$$\begin{aligned} S_T(\beta_0, \Lambda_0) &= \sqrt{T} \langle \mathbf{M}_0 \mathbf{y}_{\beta_0}^\circ, \mathbf{M}_0 \mathbf{w} \rangle_T \\ &= T^{-1/2} (\mathbf{M}_0 \mathbf{u} + \mathbf{M}_0 \mathbf{r}_0)' \mathbf{v}_0. \end{aligned}$$

Because $\mathbf{M}_0$ is symmetric and idempotent and $\mathbf{v}_0 = \mathbf{M}_0 \mathbf{w}$, we have $\mathbf{M}_0 \mathbf{v}_0 = \mathbf{v}_0$. Hence

$$(\mathbf{M}_0 \mathbf{u})' \mathbf{v}_0 = \mathbf{u}' \mathbf{v}_0, \quad (\mathbf{M}_0 \mathbf{r}_0)' \mathbf{v}_0 = \mathbf{r}_0' \mathbf{v}_0.$$

Substituting these two identities gives (5.1). The term $T^{-1/2} \mathbf{u}' \mathbf{v}_0$ is handled by the score CLT and variance consistency. The term $T^{-1/2} \mathbf{r}_0' \mathbf{v}_0$ is the only approximation-rate term in this decomposition, because it is the only term containing the sieve residual $\mathbf{r}_0$. $\square$

6 Omitted Breaks: False Predictivity as a Projection Drift

Let $\mathcal{N}_{K, 0, T}$ be the stable/no-break nuisance space and let $\mathbf{M}_{\text{st}}$ be the residual-maker for orthogonal projection onto $\mathcal{N}_{K, 0, T}$. Define

$$\mathbf{v}_{\text{st}} = \mathbf{M}_{\text{st}} \mathbf{w}, \quad S_T^{\text{st}}(\beta) = \sqrt{T} \langle \mathbf{M}_{\text{st}}(\mathbf{y} - \beta \mathbf{x}), \mathbf{M}_{\text{st}} \mathbf{w} \rangle_T.$$

Proposition 1 (Omitted-break projection drift). *At the true parameter β_0 ,*

$$S_T^{\text{st}}(\beta_0) = T^{-1/2} \mathbf{u}' \mathbf{v}_{\text{st}} + B_T, \quad B_T = T^{-1/2} \boldsymbol{\mu}'_0 \mathbf{v}_{\text{st}}. \quad (6.1)$$

Equivalently,

$$B_T = \sqrt{T} \langle \mathbf{M}_{\text{st}} \boldsymbol{\mu}_0, \mathbf{M}_{\text{st}} \mathbf{w} \rangle_T. \quad (6.2)$$

If

$$\langle \mathbf{M}_{\text{st}} \boldsymbol{\mu}_0, \mathbf{M}_{\text{st}} \mathbf{w} \rangle_T \rightarrow b \neq 0,$$

then $B_T/\sqrt{T} \rightarrow b$, so the stable score has deterministic drift of order $\sqrt{T}$ at the true parameter.

Proof. At β_0 ,

$$\mathbf{y} - \beta_0 \mathbf{x} = \boldsymbol{\mu}_0 + \mathbf{u}.$$

Applying the stable residual-maker gives

$$\mathbf{M}_{\text{st}}(\mathbf{y} - \beta_0 \mathbf{x}) = \mathbf{M}_{\text{st}} \mathbf{u} + \mathbf{M}_{\text{st}} \boldsymbol{\mu}_0.$$

Therefore

$$\begin{aligned} S_T^{\text{st}}(\beta_0) &= T^{-1/2} (\mathbf{M}_{\text{st}} \mathbf{u} + \mathbf{M}_{\text{st}} \boldsymbol{\mu}_0)' \mathbf{v}_{\text{st}} \\ &= T^{-1/2} \mathbf{u}' \mathbf{v}_{\text{st}} + T^{-1/2} \boldsymbol{\mu}'_0 \mathbf{v}_{\text{st}}, \end{aligned}$$

because $\mathbf{M}_{\text{st}}$ is symmetric and idempotent and $\mathbf{v}_{\text{st}} = \mathbf{M}_{\text{st}} \mathbf{w}$. This proves (6.1). The same symmetry and idempotence give

$$T^{-1/2} \boldsymbol{\mu}'_0 \mathbf{v}_{\text{st}} = T^{-1/2} \boldsymbol{\mu}'_0 \mathbf{M}_{\text{st}} \mathbf{w} = \sqrt{T} \langle \mathbf{M}_{\text{st}} \boldsymbol{\mu}_0, \mathbf{M}_{\text{st}} \mathbf{w} \rangle_T,$$

which proves (6.2). If the empirical inner product converges to $b \neq 0$, the drift is $B_T = \sqrt{T} b + o(\sqrt{T})$. $\square$

As a toy example, suppose there are no covariates and the stable nuisance space contains only constants:

$$\mathcal{N}_{K,0,T} = \text{span}\{\mathbf{1}_T\}.$$

Let the true nuisance contain one level shift,

$$\mu_{0,t} = \delta \mathbf{1}\{t > k_0\}, \quad \pi_T = T^{-1} \sum_{t=1}^T \mathbf{1}\{t > k_0\}.$$

The stable projection removes only the global mean. Hence

$$(\mathbf{M}_{\text{st}} \boldsymbol{\mu}_0)_t = \delta \{\mathbf{1}(t > k_0) - \pi_T\}.$$

If the residualized score direction is aligned with this step, for instance $\mathbf{M}_{\text{st}} \mathbf{w} = \mathbf{M}_{\text{st}} \boldsymbol{\mu}_0$, then

$$\langle \mathbf{M}_{\text{st}} \boldsymbol{\mu}_0, \mathbf{M}_{\text{st}} \mathbf{w} \rangle_T = \|\mathbf{M}_{\text{st}} \boldsymbol{\mu}_0\|_T^2 = \delta^2 \pi_T (1 - \pi_T).$$

When $\delta \neq 0$ and $\pi_T \rightarrow \pi \in (0, 1)$, the limit is $\delta^2 \pi (1 - \pi) \neq 0$. The stable score therefore reports a false signal even at β_0 .

The correct break-aware block sieve changes the projection target. Under the true partition,

$$\boldsymbol{\mu}_0 = \mathbf{Q}_0 \mathbf{c}_0 + \mathbf{r}_0, \quad \mathbf{Q}_0 \mathbf{c}_0 \in \mathcal{N}_{K, \Lambda_0, T}, \quad \|\mathbf{r}_0\|_T = o_p(1),$$

so $\mathbf{M}_0 \boldsymbol{\mu}_0 = \mathbf{r}_0$ is small. Under the stable space, $\mathbf{M}_{\text{st}} \boldsymbol{\mu}_0$ need not be small. The block-sieve projection removes the low-frequency break component before testing β ; otherwise omitted nuisance breaks can enter the score as false predictivity.

7 Estimated Partition

The estimated break procedure chooses a random subspace

$$\widehat{\mathcal{N}}_T = \mathcal{N}_{K, \widehat{\Lambda}, T}, \quad \widehat{\mathbf{M}}_T = \mathbf{M}_{K, \widehat{\Lambda}}.$$

The feasible score is

$$\widehat{S}_T(\beta_0) = \sqrt{T} \langle \widehat{\mathbf{M}}_T(\mathbf{y} - \beta_0 \mathbf{x}), \widehat{\mathbf{M}}_T \mathbf{w} \rangle_T.$$

The true-partition score is

$$S_T^0(\beta_0) = \sqrt{T} \langle \mathbf{M}_0(\mathbf{y} - \beta_0 \mathbf{x}), \mathbf{M}_0 \mathbf{w} \rangle_T.$$

The estimated-partition Wilks argument is the score-equivalence statement

$$\widehat{S}_T(\beta_0) - S_T^0(\beta_0) = o_p(1), \quad \widehat{V}_T - V_T^0 = o_p(1). \quad (7.1)$$

Geometrically, (7.1) says that replacing the true subspace $\mathcal{N}_{K, \Lambda_0, T}$ by the estimated subspace $\mathcal{N}_{K, \widehat{\Lambda}, T}$ changes the score only by a negligible amount at the $T^{-1/2}$ scale.

Remark 1 (Why this is weaker than operator-norm convergence). It is not necessary to prove $\|\widehat{\mathbf{M}}_T - \mathbf{M}_0\|_{op} \rightarrow 0$. Break-date errors can make that hard. The theorem only needs the projected score and its second moment to be asymptotically unchanged.

8 EL as a Finite-Sample Convex KKT Problem

For fixed (K, Λ, β) , everything in this section is finite-sample algebra in $L^2(P_T)$. Define the profiled residual, score direction, and scalar moment array

$$\mathbf{r}_\beta = \mathbf{M}_{K, \Lambda}(\mathbf{y} - \beta \mathbf{x}), \quad \mathbf{a} = \mathbf{M}_{K, \Lambda} \mathbf{w}, \quad g_t(\beta, \Lambda) = r_{\beta, t} a_t.$$

Then

$$\Psi_T(\beta, \Lambda) = P_T g(\beta, \Lambda) = T^{-1} \sum_{t=1}^T g_t(\beta, \Lambda) = \langle \mathbf{r}_\beta, \mathbf{a} \rangle_T.$$

Thus the sieve basis $Q_{K, \Lambda}$ has already represented the nuisance feasible directions: $\text{col}(Q_{K, \Lambda})$ is the tangent plane for nuisance variation, and $\mathbf{M}_{K, \Lambda}$ keeps only the orthogonal score component.

EL adds one more finite-dimensional convex problem. The primal feasible set is the probability simplex cut by the affine moment hyperplane

$$\mathcal{C}_T(\beta, \Lambda) = \left\{ \pi \in \mathbb{R}_+^T : \sum_{t=1}^T \pi_t = 1, \quad \sum_{t=1}^T \pi_t g_t(\beta, \Lambda) = 0 \right\}.$$

The empirical likelihood problem is equivalently

$$\min_{\pi \in \mathcal{C}_T(\beta, \Lambda)} - \sum_{t=1}^T \log(T\pi_t). \quad (8.1)$$

This is a convex barrier objective over a convex feasible set. Its Lagrangian is

$$\mathcal{L}(\pi, \eta, \lambda) = - \sum_{t=1}^T \log(T\pi_t) + \eta \left(\sum_{t=1}^T \pi_t - 1 \right) + T\lambda \sum_{t=1}^T \pi_t g_t.$$

The KKT first-order condition gives

$$-\frac{1}{\pi_t} + \eta + T\lambda g_t = 0.$$

After normalizing by $\sum_t \pi_t = 1$, the EL weights are

$$\hat{\pi}_t(\lambda) = \frac{1}{T\{1 + \lambda g_t(\beta, \Lambda)\}}, \quad 1 + \lambda g_t(\beta, \Lambda) > 0. \quad (8.2)$$

The remaining KKT equation is the dual moment equation

$$P_T \left[\frac{g_t(\beta, \Lambda)}{1 + \lambda g_t(\beta, \Lambda)} \right] = 0. \quad (8.3)$$

The log empirical likelihood ratio is therefore

$$\ell_T(\beta, \Lambda) = 2 \sum_{t=1}^T \log\{1 + \lambda_T g_t(\beta, \Lambda)\}. \quad (8.4)$$

The quadratic EL reduction is also mostly KKT algebra. If $\max_t |\lambda_T g_t| = o_p(1)$, then expanding (8.3) gives

$$0 = P_T \left[\frac{g_t}{1 + \lambda_T g_t} \right] = P_T g_t - \lambda_T P_T g_t^2 + R_{1T}, \quad R_{1T} = o_p(T^{-1/2}).$$

Hence, with $V_T = P_T g_t^2$,

$$\lambda_T = \frac{P_T g_t}{V_T} + o_p(T^{-1/2}).$$

Expanding (8.4) then yields

$$\ell_T(\beta_0, \Lambda) = \frac{T\{P_T g_t(\beta_0, \Lambda)\}^2}{V_T(\Lambda)} + o_p(1) = \frac{S_T(\beta_0, \Lambda)^2}{V_T(\Lambda)} + o_p(1). \quad (8.5)$$

The stochastic work is not to rediscover EL; it is only to verify the small size conditions under which the KKT expansion is valid.

9 Population Hilbert Space

9.1 Population object in $L^2(P)$

To express breaks in population geometry, include rescaled time $S \in (0, 1]$. Let

$$O = (Y, X, W, S, U)$$

be a population random element with law P . The true regime map is

$$j_0(S) = j \iff \tau_{j,0} < S \leq \tau_{j+1,0}.$$

The population model is the identity in $L^2(P)$:

$$Y = \beta_0 X + \mu_0(S, W) + U, \quad \mu_0(S, W) = \sum_{j=0}^{q_0} m_j(W) \mathbf{1}\{\tau_{j,0} < S \leq \tau_{j+1,0}\}. \quad (9.1)$$

9.2 Population nuisance space

For a population partition Λ , define

$$\mathcal{N}_\Lambda^P = \overline{\left\{ \sum_{j=0}^q \mathbf{1}\{S \in I_j(\Lambda)\} h_j(W) : h_j \in L^2(P_W) \right\}}^{L^2(P)}.$$

Let Π_Λ^P be the $L^2(P)$ -orthogonal projection onto $\mathcal{N}_\Lambda^P$, and let

$$M_\Lambda^P = I - \Pi_\Lambda^P.$$

For $\beta \in \mathcal{B}$, define the population partial outcome

$$Y_\beta^\circ = Y - \beta X.$$

The population profiled residual and score direction are

$$r_\beta^P(\Lambda) = M_\Lambda^P Y_\beta^\circ = M_\Lambda^P (Y - \beta X), \quad v^P(\Lambda) = M_\Lambda^P w.$$

The population moment is

$$\Psi(\beta, \Lambda) = \langle r_\beta^P(\Lambda), v^P(\Lambda) \rangle_P. \quad (9.2)$$

10 Population Identification

At the true population partition Λ_0 , abbreviate

$$\mathcal{N}_0^P = \mathcal{N}_{\Lambda_0}^P, \quad \Pi_0^P = \Pi_{\Lambda_0}^P, \quad M_0^P = M_{\Lambda_0}^P, \quad v_0^P = v^P(\Lambda_0) = M_0^P w.$$

Because the true nuisance is regime-specific with the true population partition,

$$\mu_0(S, W) \in \mathcal{N}_0^P.$$

Therefore the true-partition population residual-maker removes it:

$$M_0^P \mu_0 = 0. \quad (10.1)$$

This is the $L^2(P)$ analog of the finite-sample statement that the correct break-aware nuisance space absorbs the nuisance component.

Using the population model (9.1), the population partial outcome at a generic β is

$$\begin{aligned} Y_\beta^\circ &= Y - \beta X \\ &= \mu_0(S, W) + U - (\beta - \beta_0)X. \end{aligned}$$

Projecting onto $(\mathcal{N}_0^P)^\perp$ gives

$$\begin{aligned} r_\beta^P(\Lambda_0) &= M_0^P Y_\beta^\circ \\ &= M_0^P \mu_0 + M_0^P U - (\beta - \beta_0) M_0^P X \\ &= M_0^P U - (\beta - \beta_0) M_0^P X, \end{aligned} \quad (10.2)$$

because $M_0^P \mu_0 = 0$. Taking the inner product with the true population score direction v_0^P gives

$$\Psi(\beta, \Lambda_0) = \langle M_0^P U, v_0^P \rangle_P - (\beta - \beta_0) \langle M_0^P X, v_0^P \rangle_P. \quad (10.3)$$

This derivation is algebraic. The two population assumptions enter only after (10.3) is obtained.

The exogeneity condition is

$$\langle M_0^P U, v_0^P \rangle_P = 0.$$

The relevance condition is

$$\Gamma = \langle M_0^P X, v_0^P \rangle_P \neq 0.$$

Under these two conditions,

$$\Psi(\beta, \Lambda_0) = -(\beta - \beta_0)\Gamma.$$

Thus

$$\Psi(\beta, \Lambda_0) = 0 \iff \beta = \beta_0. \quad (10.4)$$

Part II

Small Stochastic Bounds

11 Small Stochastic Bounds

After Part I, the finite-sample objects are fixed: the nuisance space is $\text{col}(Q_{K,\Lambda})$, profiling is $M_{K,\Lambda}$, the score is $g_t = r_{\beta,t} a_t$, and EL is the KKT system (8.2)–(8.3). The stochastic part should therefore be small. It only verifies that these finite-sample objects are close to their population limits and that the KKT expansion is used in a valid local neighborhood.

The first stochastic target is the uniform moment bridge

$$\sup_{\beta \in \mathcal{B}} |\Psi_T(\beta, \hat{\Lambda}) - \Psi(\beta, \Lambda_0)| = o_p(1). \quad (11.1)$$

This is the stochastic statement that the empirical projected inner product converges to the population projected inner product.

A clean lemma list is:

1. **Gram stability.** The empirical Gram matrices for $Q_{K,\Lambda}$, x , and w converge uniformly over admissible partitions.
2. **Sieve approximation.** Lemma 1 states the desired uniform $L^2(P_T)$ approximation. The stochastic task is to verify $\rho_{K,T}(\mathcal{H}, \Lambda_0) = o_p(1)$, so the residual for the true nuisance, $M_{K,\Lambda_0}\mu_0$, is negligible in the required score norm.
3. **Break localization.** Replacing Λ_0 by $\hat{\Lambda}$ changes the projected score only by $o_p(1)$ at the moment scale and by $o_p(1)$ at the $T^{-1/2}$ score scale when Wilks is the target.
4. **Uniform LLN.** The projected sample moment obeys (11.1).
5. **Score CLT.** At the true value,

$$S_T(\beta_0, \hat{\Lambda}) = \sqrt{T} \Psi_T(\beta_0, \hat{\Lambda}) \Rightarrow N(0, V).$$

6. **Variance consistency.** With $V_T(\hat{\Lambda}) = P_T g_t(\beta_0, \hat{\Lambda})^2$,

$$V_T(\hat{\Lambda}) \rightarrow_p V > 0.$$

7. **EL local feasibility.** The origin lies in the interior of the convex hull of $\{g_t(\beta_0, \hat{\Lambda})\}_{t=1}^T$ with high probability, and

$$\max_{t \leq T} |g_t(\beta_0, \hat{\Lambda})|/\sqrt{T} = o_p(1).$$

These conditions make $1 + \lambda_T g_t > 0$ and justify the Taylor reduction in (8.5).

If $\hat{\beta}_T$ is an approximate sample root,

$$\Psi_T(\hat{\beta}_T, \hat{\Lambda}) = o_p(1),$$

then (11.1) gives

$$\Psi(\hat{\beta}_T, \Lambda_0) = o_p(1).$$

By unique population identification, $\hat{\beta}_T \rightarrow_p \beta_0$. This is where the stochastic bridge feeds the geometric identification result.

Part III

Merging Part

12 Merging Part: Moment Bridge, EL KKT, and χ_1^2

The merge step should not say that the unscaled moment bridge is itself equivalent to Wilks. The correct statement is more precise:

The same projected moment Ψ_T is the interface for both targets. Unscaled convergence of Ψ_T to Ψ merges with population uniqueness to prove consistency. Scaled convergence of Ψ_T , together with the EL KKT quadratic reduction, merges to prove χ_1^2 .

12.1 Merge A: consistency

Part I gives the population geometry

$$\Psi(\beta, \Lambda_0) = \langle M_0^P Y_\beta^\circ, v_0^P \rangle_P = -(\beta - \beta_0)\Gamma$$

under exogeneity and relevance. Hence

$$\Psi(\beta, \Lambda_0) = 0 \iff \beta = \beta_0.$$

Part II gives the stochastic bridge

$$\sup_{\beta \in \mathcal{B}} |\Psi_T(\beta, \hat{\Lambda}) - \Psi(\beta, \Lambda_0)| = o_p(1).$$

Therefore any approximate sample root satisfies

$$\Psi_T(\hat{\beta}_T, \hat{\Lambda}) = o_p(1) \implies \hat{\beta}_T \rightarrow_p \beta_0.$$

This is the estimation target.

12.2 Merge B: Wilks inference

For inference, Part I gives the EL KKT representation

$$\ell_T(\beta, \hat{\Lambda}) = 2 \sum_{t=1}^T \log\{1 + \lambda_T g_t(\beta, \hat{\Lambda})\},$$

where λ_T solves

$$P_T \left[\frac{g_t(\beta, \hat{\Lambda})}{1 + \lambda_T g_t(\beta, \hat{\Lambda})} \right] = 0.$$

Part II verifies the local size and convex-hull conditions, so the KKT equation can be expanded at $\beta = \beta_0$:

$$\ell_T(\beta_0, \hat{\Lambda}) = \frac{T\{\Psi_T(\beta_0, \hat{\Lambda})\}^2}{V_T(\hat{\Lambda})} + o_p(1).$$

Part II also gives

$$\frac{\sqrt{T}\Psi_T(\beta_0, \hat{\Lambda})}{\sqrt{V_T(\hat{\Lambda})}} \Rightarrow N(0, 1).$$

Combining the last two displays gives

$$\ell_T(\beta_0, \hat{\Lambda}) \Rightarrow \chi_1^2. \tag{12.1}$$

Thus the “equivalence” is conditional on the geometric KKT reduction. Once EL has been reduced to the square of the standardized projected moment, Wilks is equivalent to the standardized score CLT. It is not equivalent to the unscaled LLN bridge used for consistency.

Proof-body split. Geometry owns the model, nuisance tangent space, projection KKT, residual score, population zero, and EL primal/dual KKT. Small stochastic bounds only verify convergence and local size. The merging part then turns the same projected moment into either consistency or Wilks.

13 Three-Part Logic Checklist

Proof part	Object	What it proves
Geometry / KKT	$\mathbf{y} = \beta_0 \mathbf{x} + \boldsymbol{\mu}_0 + \mathbf{u} \in L^2(P_T)$	The full sample model is a Hilbert-space identity.
Geometry / KKT	$\mathcal{N}_{K,\Lambda,T} = \text{col}(Q_{K,\Lambda})$	The sieve basis represents the nuisance tangent plane.
Geometry / KKT	$M_{K,\Lambda} = I - \Pi_{K,\Lambda,T}$	Profiling is the KKT normal equation for orthogonal projection.
Geometry / KKT	$g_t = \{M(y - \beta x)\}_t \{Mw\}_t$	The profile score is one projected inner product.
Geometry / KKT	EL primal/dual KKT	EL is a convex simplex problem cut by the moment hyperplane.
Population geometry	$\Psi(\beta, \Lambda_0) = -(\beta - \beta_0)\Gamma$	Exogeneity and relevance give the unique population zero.
Small stochastic	$\sup_{\beta} \Psi_T - \Psi = o_p(1)$	The sample projected moment reaches the population moment.
Small stochastic	CLT, variance, max-score, convex hull	The EL KKT expansion is valid locally and the standardized score is normal.
Merging	Moment bridge plus unique zero	Consistency: $\hat{\beta}_T \rightarrow_p \beta_0$.
Merging	EL KKT quadratic reduction plus CLT	Wilks: $\ell_T(\beta_0, \hat{\Lambda}) \Rightarrow \chi_1^2$.

14 Minimal Final Statement

The full profile-sieve EL proof is a three-part argument.

Geometric/KKT body. In $L^2(P_T)$, the model is $\mathbf{y} = \beta_0 \mathbf{x} + \boldsymbol{\mu}_0 + \mathbf{u}$. A candidate partition and sieve define the nuisance tangent plane $\mathcal{N}_{K,\Lambda,T} = \text{col}(Q_{K,\Lambda})$. Profiling is orthogonal projection onto this space. The residual score is the projected inner product

$$\Psi_T(\beta, \Lambda) = \langle M_{K,\Lambda}(\mathbf{y} - \beta \mathbf{x}), M_{K,\Lambda} \mathbf{w} \rangle_T.$$

EL is then a convex simplex problem with the moment hyperplane $\sum_t \pi_t g_t = 0$, whose KKT system gives $\pi_t = 1/[T\{1 + \lambda g_t\}]$.

Small stochastic body. The stochastic lemmas verify that the empirical projection, estimated break partition, sieve approximation, score variance, convex

hull, and max-score terms are small enough for the sample KKT objects to approximate their population limits.

Merging body. For consistency, combine $\Psi_T \rightarrow \Psi$ with the unique population zero. For Wilks, combine the EL KKT quadratic reduction with the standardized score CLT:

$$\ell_T(\beta_0, \hat{\Lambda}) = \frac{T\Psi_T(\beta_0, \hat{\Lambda})^2}{V_T(\hat{\Lambda})} + o_p(1), \quad \frac{\sqrt{T}\Psi_T(\beta_0, \hat{\Lambda})}{\sqrt{V_T(\hat{\Lambda})}} \Rightarrow N(0, 1).$$

Therefore $\ell_T(\beta_0, \hat{\Lambda}) \Rightarrow \chi_1^2$.